

Eesha Iyer

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Education

University of Maryland

Master of Science, Applied Economics

Related Coursework: Empirical Research, Microeconomic Analysis, Program Analysis & Evaluation

College Park, MD
Expected May 2025

Indian Institute of Technology, Madras

Postgraduate Diploma in Data Science

Chennai, India
2021-2023

University of Madras

Bachelor of Arts, Economics

Chennai, India
2019-2022

Experience

The World Bank Group

Global Indicators Group | Research Intern

Washington, D.C.
Sept 2024 – Dec 2024

- Built a Python-based DOCX-to-XLSX mapping system using openpyxl for cross-lingual data extraction, ensuring 82% accuracy
- Analyzed 5,000+ data points from stakeholder surveys using clustering and regression analysis to assess policy impact
- Developed Python scripts to preprocess survey data and visualize historical project data for a published World Bank paper

University of Maryland

Center for Educational Data Science and Innovation | Research Assistant

College Park, MD
Jan 2025 – Present

- Built predictive models using STATA to assess the impact of education policy changes, improving forecast accuracy by 12%
- Used value-added models (VAM) to conduct impact analysis of actions on student infractions across 100+ schools
- Created data visualizations using PowerBI and Matplotlib to deliver insights to county and district stakeholders

UMB CURE Program, College of Education | Research Assistant

Feb 2024 – Sept 2024

- Conducted statistical analysis on 50+ school datasets using STATA to assess the impact of STEM programs on student outcomes
- Cleaned and merged complex datasets using STATA, reducing data processing time by 30%

Robert H. Smith School of Business | Research Assistant

Feb 2024 – May 2024

- Analyzed healthcare market data using Python and R to assess industry trends and ownership structures
- Developed a regression model to quantify the impact of technology adoption on healthcare outcomes
- Synthesized research findings into presentations using PowerBI, enhancing stakeholder engagement

TVS Credit Services Ltd

Data & Analytics Intern

Chennai, India
May 2023 - Jul 2023

- Utilized diverse survey methods to gather datasets essential for algorithm training, ensuring coverage of real-world scenarios
- Achieved 97% accuracy in vehicle license plate recognition Optical Character Recognition Model using YOLOv3
- Performed demographic analysis using Python (Pandas, NumPy) to identify patterns among first-time financial product users

Projects

Green Space Accessibility and Population Analysis | Python | [\[Link\]](#)

Developed an interactive geospatial analysis tool to assess urban green space accessibility. Combined city boundary and park data with raster population datasets to identify underserved areas through buffer analysis and visualization.

Skills: Geospatial Analysis, Raster Data Processing, Geospatial Visualization, Shapely, GeoPandas, Folium

S&P 500 AI-Powered Portfolio Optimization | Python | [\[Link\]](#)

Built a Python-based AI agent using Modern Portfolio Theory and Deep Q-Networks to dynamically optimize S&P 500 portfolios; deployed as a Streamlit app for user-input analysis and predictions.

Skills: Portfolio Optimization, Reinforcement Learning, TensorFlow, Streamlit, yfinance, scikit-learn, pandas, numpy

Skills

Programming: Python (Pandas, NumPy, scikit-learn), R, STATA, SQL, Mathematica, LaTeX

Machine Learning: Regression, SVM, PSM, Neural Networks, Random Forest, Clustering, PCA, YOLOv3, Deep Q-Networks (DQN)

Data Analysis: Data Cleaning, Visualization (Matplotlib, Seaborn, PowerBI, Excel), Time Series (ARIMA, VAR)

Certifications: Google Data Analytics, Microsoft Azure Fundamentals, Python for Everybody Specialization